Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 1 of 7 · Question 4 part b · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number				
4	(a) (i)	Two from: • no red blood cells; • no plasma/large proteins; • higher oxygen concentration in the blood; • lower carbon dioxide concentration in the blood;	ORA	2
	(ii)	• artery end has a higher pressure; • capillaries have holes/pores/are permeable to small molecules/are leaky; • water forced/passes out; • small molecules are forced out/filtered/larger molecules/named larger molecule remain in the blood;	Allow blood enters capillaries under high pressure	4
	(iii)	• excess tissue fluid drains through the lymph; • muscle contraction moves fluid in the lymphatic system;		2
	(b)	Three from: • antibodies have a variable region/contain receptors; • variable region/receptors have a specific shape/all antibodies are different; • variable region/receptors are a complementary shape to antigen; • antibodies bind to an antigen;		3

Total 11 marks

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 1 of 7 · Question 4 part b · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 2 of 7 · Question 6 part a.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 2 of 7 · Question 6 part a.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 3 of 7 · Question 7 part b.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

Number				
7	(a)	Any three from • wall is one cell thick for faster diffusion/short diffusion distance; • close network of blood vessels to decrease diffusion distance/maintain gradient; • contains microvilli to increase surface area (for absorption/diffusion); • contains a lacteal/lymph vessels to absorb fat/fatty acids/glycerol;	Adaptation and explanation needed for each mark	3
	(b) (i)	Any two from • (immune system) does not recognise self-antigens/sees self-antigens as foreign; • white blood cells/immune system produce antibodies; • antibodies bind to (self) antigens;		2
	(ii)	• villi damaged/destroyed/less villi; • decreased surface area; • less/slower absorption of nutrients/named nutrient;		3

Total for Question 7 = 8 Marks

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 4 of 7 · Question 7 part d · 2 marks

2025 JUN · Paper 02 (regional R) · find part d in this mark scheme extract

7	(a)	amylase;		1
	(b)	Any two from • enzymes are immobilised on testing strip; • react with/breaks down glucose (in urine); • (catalyse a chemical reaction that) causes a colour change (in strip);		2
	(c)	pancreas;		1
	(d)	• rejection/immune response/allergies; • (immune system) recognised it as foreign/antigen;		2
	(e)	any two from • use restriction enzymes; • cut out insulin gene from human DNA; • insert human insulin gene into yeast genome/plasmid/genetic material;		2
			Total for Question 7 = 8 marks	

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 5 of 7 · Question 8 part a.iv · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.iv in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 6 of 7 · Question 8 part a.v · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.v in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	

Type MS · Explain antigen-antibody specificity and how monoclonal antibodies are made

Answer 7 of 7 · Question 8 part whole · 5 marks

2020 JAN · Paper 02 · find part whole in this mark scheme extract

Question number			
8	• antigen injected into a mouse/other named animal(1) • spleen cells produce antibodies(1) • spleen cells are fused with tumour/myeloma cells(1) • hybridoma cells formed(1) • hybridoma cells produce monoclonal antibodies(1)	Allow lymphocytes from spleen cells produce antibodies Allow cancer cells for myeloma Allow lymphocytes from spleen cells fused	5

Total for Question 8 = 5 marks